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In[ ]:= Clear["Global`*"]

In[ ]:= (*Base Cases*)
win[4, _] = 1;
win[_ , 3] = 0;

In[ ]:= Pstrike[b_, s_] := Pstrike[b, s] = 
$$\frac{\text{win}[b, 1 + s] - \text{win}[1 + b, s]}{-4 p + \text{win}[b, 1 + s] + p \text{win}[b, 1 + s] - \text{win}[1 + b, s]}$$
;

In[ ]:= Pball[b_, s_] := Pball[b, s] = 1 - Pstrike[b, s];
Pswing[b_, s_] := Pstrike[b, s];

In[ ]:= Pwait[b_, s_] := Pball[b, s];

In[ ]:= win[b_, s_] := win[b, s] =
Simplify[Pstrike[b, s] (4 p + (1 - p) win[b, s + 1]) + (1 - Pstrike[b, s]) win[b, s + 1]];

In[ ]:= (*Base Cases for probability at vertex*)

In[ ]:= P[0, 0] = 1;
P[_ , -1] = 0;
P[-1, _] = 0;

In[ ]:= P[b_, s_] := P[b, s] = Simplify[P[b - 1, s] (Pball[b - 1, s] * Pwait[b - 1, s]) +
P[b, s - 1] (Pball[b, s - 1] * Pswing[b, s - 1] +
Pstrike[b, s - 1] * Pwait[b, s - 1] + Pstrike[b, s - 1] * Pswing[b, s - 1] (1 - p))];

In[ ]:= NumberForm[N[Maximize[{P[3, 2], 0 ≤ p ≤ 1}, p], 10]]

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Out[ ]//NumberForm=
{0.2959679934, {p → 0.2269732325}}

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In[ ]:= Plot[P[3, 2], {p, 0, 1}]

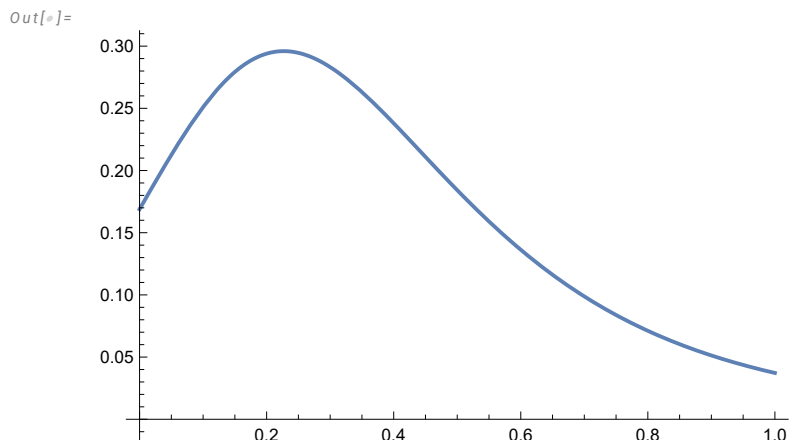
```

General: 0.0000204286⁶⁶ is too small to represent as a normalized machine number; precision may be lost. [i](#)

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General: Further output of General::munfl will be suppressed during this calculation. [i](#)



(*Solve for strike probability at
indifference and swing probability (they're the same)*)

In[*]:= Solve[a (4 p + (1 - p) w[b, s + 1]) + (1 - a) w[b, s + 1] == a w[b, s + 1] + (1 - a) w[b + 1, s], a]

Out[*]=

$$\left\{ \left\{ a \rightarrow \frac{w[b, 1 + s] - w[1 + b, s]}{-4 p + w[b, 1 + s] + p w[b, 1 + s] - w[1 + b, s]} \right\} \right\}$$

In[*]:= Solve[sw (4 p + (1 - p) w[b, s + 1]) + (1 - sw) w[b, s + 1] == sw w[b, s + 1] + (1 - sw) w[b + 1, s], sw]

Out[*]=

$$\left\{ \left\{ sw \rightarrow \frac{w[b, 1 + s] - w[1 + b, s]}{-4 p + w[b, 1 + s] + p w[b, 1 + s] - w[1 + b, s]} \right\} \right\}$$